SUMMARY

- If $y = f(x)$, then $\frac{dy}{dx}$ represents instantaneous rate of change of y with respect to x.
- If $y = f\left(g(t)\right)$, then $\frac{dy}{dt} = f'\left(g(t)\right) \cdot g'(t)$ which is called the chain rule.
- The equation of tangent at (a,b) to the curve $y = f(x)$ is given by $y - b = \left(\frac{dy}{dx}\right)_{(a,b)} (x-a)$ or $y - b = f'(a)(x-a)$.
- Rolle's Theorem

  Let $f(x)$ be continuous in a closed interval $[a,b]$ and differentiable on the open interval (a,b). If $f(a) = f(b)$, then there is at least one point $c \in (a,b)$ where $f'(c) = 0$.
- Lagrange's Mean Value Theorem

  Let $f(x)$ be continuous in a closed interval $[a,b]$ and differentiable on the open interval (a,b) (where $f(a)$ and $f(b)$ are not necessarily equal). Then there is at least one point $c \in (a,b)$ such that $f'(c) = \frac{f(b) - f(a)}{b-a}$.
- Taylor's series

  Let $f(x)$ be a function infinitely differentiable at $x = a$. Then $f(x)$ can be expanded as a series in an interval $(a-\varepsilon, a+\varepsilon), \varepsilon > 0$, of the form

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{\lfloor n} (x-a)^n = f(a) + \frac{f'(a)}{\lfloor 1}(x-a) + \cdots + \frac{f^{(n)}(a)}{\lfloor n}(x-a)^n + \cdots$$

- Maclaurin's series

  In the Taylor's series if $a = 0$, then the expansion takes the form

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{\lfloor n} (x)^n = f(0) + \frac{f'(0)}{\lfloor 1}(x) + \cdots + \frac{f^{(n)}(0)}{\lfloor n}(x^n) + \cdots$$

- The l'Hôpital's rule

  Suppose $f(x)$ and $g(x)$ are differentiable functions and $g'(x) \neq 0$ with

  $\lim_{x \to a} f(x) = 0 = \lim_{x \to a} g(x)$. Then $\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}$

  $\lim_{x \to a} f(x) = \pm\infty = \lim_{x \to a} g(x)$. Then $\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}$
- If the function $f(x)$ is differentiable in an open interval (a,b) then we say, if $\frac{d}{dx}\left(f(x)\right) > 0$, $\forall x \in (a,b)$ then $f(x)$ is strictly increasing in the interval (a,b).

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